

2)Skill

A)

```
GNU nano 8.7.1 Skill-2.c
#include <stdio.h>
#include <string.h>

int main() {
    char input[100];

    while (1) {
        printf("myshell> ");
        fgets(input, sizeof(input), stdin);

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0) {
            break;
        }

        printf("You entered: %s\n", input);
    }

    printf("Shell exited.\n");
    return 0;
}
```

A) Output:

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill$ cd Skill-02
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-02$ nano Skill-2.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-02$ gcc Skill-2.c -o Skill-2
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-02$ ./Skill-2
myshell> helo
You entered: helo
myshell> hello
You entered: hello
myshell> exit
Shell exited.
```

B) Code Main Loop:

```
GNU nano 8.7.1 MainLoop.c
#include <stdio.h>
#include <unistd.h>
#include <termios.h>
#include <string.h>

int main() {
    char buffer[100];
    int index = 0;
    char ch;

    struct termios oldt, newt;

    tcgetattr(STDIN_FILENO, &oldt);
    newt = oldt;

    newt.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSANOW, &newt);

    printf("myshell> ");

    while (1) {
        ch = getchar();

        if (ch == '\n') {
            buffer[index] = '\0';

            printf("\nYou entered: %s\n", buffer);
            break;
        }

        else if (ch == 127) {
            if (index > 0) {
                index--;
                printf("\b \b");
                fflush(stdout);
            }
        }

        else if (index < 99) {
            buffer[index] = ch;
            index++;
            printf("%c", ch);
            fflush(stdout);
        }
    }

    tcsetattr(STDIN_FILENO, TCSANOW, &oldt);

    return 0;
}
```

Output:

B)

```
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-02$ nano MainLoop.c
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-02$ gcc MainLoop.c -o MainLoop
uday@DESKTOP-47L1GTF:~/2520090089_OSSP/2520090089_Skill/Skill-02$ ./MainLoop
myshell> hello
You entered: hello
```